Partial Tax Recovery on Returns
(α-Reclaim under Seller and Buyer Remittance)

Draft notes
Companion to `welfare_with_returns`

August 14, 2026

1 Setup

The companion note studies the extreme in which the remitter recovers none of the unit tax τ from the government when a unit is returned. Here the remitter recovers a fraction $\alpha \in [0, 1)$ of τ on each returned unit, whether the seller or the buyer remits. Full reclaim is the boundary $\alpha = 1$; zero reclaim is $\alpha = 0$. Private refunds between buyer and seller are unchanged from the companion note. Under seller remittance the seller refunds the tax-inclusive checkout price p. Under buyer remittance the seller refunds only the pre-tax list price $p - \tau$, and the buyer reclaims $\alpha\tau$ from the treasury on a return.

Demand $D(p)$, return rate $r(\cdot)$, handling h, and market supply $S(\mu)$ are as in the companion note. Equilibrium clears at $D(p) = S(\mu)$, with μ the remitter- and α-dependent effective margin derived below. Write $Q \equiv D(p)$.

2 Effective margins

Under seller remittance the seller remits τ on every gross sale and recovers $\alpha\tau$ from the government on each return. The private refund remains p, so the return rate is still $r(p)$. Expected net receipt per gross unit is therefore

$$\mu^s(p, \tau; \alpha) = p - \tau - r(p)\big(p + h - \alpha\tau\big). \tag{1}$$

The term $-\alpha\tau$ inside the return cost is the treasury reclaim. At $\alpha = 0$ this is the companion-note margin $p - \tau - r(p)(p + h)$.

Under buyer remittance the seller never remits τ. The buyer remits on purchase and recovers $\alpha\tau$ on a return, while the seller refunds only the pre-tax price $p - \tau$. The buyer's total recovery on a return is therefore

$$(p - \tau) + \alpha\tau = p - (1 - \alpha)\tau,$$

so the return rate is evaluated at $p - (1 - \alpha)\tau$. The seller's effective margin is

$$\mu^b(p, \tau; \alpha) = (p - \tau) - r\big(p - (1 - \alpha)\tau\big)\,(p - \tau + h). \tag{2}$$

At $\alpha = 0$ the argument of r collapses to $p - \tau$ and (2) recovers the companion note. Clearing conditions are $D(p) = S\big(\mu^s(p, \tau; \alpha)\big)$ and $D(p) = S\big(\mu^b(p, \tau; \alpha)\big)$.

3 Tax neutrality

At a common checkout price p and tax τ, the two regimes define the same equilibrium if and only if $\mu^s = \mu^b$. Canceling $p - \tau$ gives

$$r(p)\big(p + h - \alpha\tau\big) = r\big(p - (1-\alpha)\tau\big)\,(p - \tau + h). \tag{3}$$

Result 1 (Partial recovery and tax neutrality)**.** *Seller and buyer remittance are equivalent if and only if (3) holds. When $\alpha = 1$, both sides equal $r(p)(p - \tau + h)$, so neutrality holds for every r. When $\alpha \in [0, 1)$, neutrality fails for generic r, as in the zero-reclaim companion note.*

Full reclaim undoes the remittance wedge on returns. Any incomplete reclaim leaves a wedge between the seller's return cost and the buyer's recovery on a return, so which party remits continues to matter.

4 Pass-through

Production-cost and handling pass-through ρ_c and ρ_h are as in the companion note and are unchanged by reclaim. What changes is the tax wedge. Failure of neutrality requires separate checkout prices p^s and p^b. With $S_\tau = 0$,

$$\rho^s_\tau = \frac{-S'(\mu)\,\mu^s_\tau}{S'(\mu)\,\mu^s_p - D'(p^s)}, \qquad \rho^b_\tau = \frac{-S'(\mu)\,\mu^b_\tau}{S'(\mu)\,\mu^b_p - D'(p^b)}. \tag{4}$$

Differentiating (1)–(2) at fixed checkout price, and writing $z \equiv p - (1-\alpha)\tau$,

$$\mu^s_p = 1 - r(p) - r'(p)\big(p + h - \alpha\tau\big), \qquad \mu^s_\tau = -(1 - \alpha r(p)), \tag{5}$$

$$\mu^b_p = 1 - r(z) - r'(z)\,(p - \tau + h), \qquad \mu^b_\tau = -\big(1 - r(z)\big) + r'(z)\,(1-\alpha)\,(p - \tau + h). \tag{6}$$

Seller remittance lowers μ by the unclaimed expected tax $1 - \alpha r(p)$. Buyer remittance lowers the buyer's recovery on a return by only $1 - \alpha$, so the return-rate channel in μ^b_τ is scaled by that factor.

At $\tau = 0$, both list-price slopes collapse to the companion object $\mu_p \equiv 1 - r(p) - r'(p)(p + h)$, and

$$\mu^s_\tau = -(1 - \alpha r(p)), \qquad \mu^b_\tau = -\big(1 - r(p) - (1-\alpha)\,r'(p)(p + h)\big). \tag{7}$$

Comparing with ρ_c then yields

$$\rho^s_\tau = (1 - \alpha r(p))\,\rho_c, \qquad \rho^b_\tau = \big(1 - r(p) - (1-\alpha)\,r'(p)(p + h)\big)\,\rho_c. \tag{8}$$

Result 2 (Tax pass-through under partial reclaim)**.** *At $\tau = 0$, tax pass-through is (8). The formulas nest the companion note at $\alpha = 0$, where $\rho^s_\tau = \rho_c$ and $\rho^b_\tau = \mu_p\rho_c$, and coincide at $(1 - r(p))\rho_c$ when $\alpha = 1$. For $\alpha \in [0, 1)$, $\operatorname{sgn}(\rho^b_\tau) = -\operatorname{sgn}(\rho^s_\tau)$ if and only if $1 - r(p) - (1-\alpha)\,r'(p)(p + h) < 0$ and $\rho_c \neq 0$.*

At $\tau = 0$, the list-price slope μ_p does not depend on reclaim, so demand and how supply responds to p are the same as for a production-cost shock. Seller-tax pass-through therefore scales only with how much τ lowers the effective margin—and thus creates excess demand—at a fixed checkout price. That shift is $1 - \alpha r(p)$: the firm remits one dollar on every gross sale but recovers α on the fraction $r(p)$ that returns, so expected unclaimed tax falls in α. Hence $\rho^s_\tau = (1 - \alpha r)\,\rho_c$, which declines from ρ_c at $\alpha = 0$ toward $(1 - r)\rho_c$ as reclaim becomes complete: more complete recovery leaves a smaller excess-demand gap, so checkout-price pass-through is weaker. Cost and handling pass-through are untouched because those shocks never enter the $\alpha\tau$ reclaim term.

Under buyer remittance, pass-through again scales only with how τ shifts the effective margin at a fixed checkout price. The buyer's recovery on a return is the seller's pre-tax refund plus the treasury reclaim, $p-(1-\alpha)\tau$. A one-dollar tax therefore lowers that recovery by $1-\alpha$, so the return-rate response is scaled by the unreclaimed share. The resulting factor is $1-r-(1-\alpha)r'(p)(p+h)$. If $r' < 0$, the unreclaimed tax raises the return rate and enlarges the hit to the effective margin above $1-r$. More complete reclaim shrinks that channel, so the factor falls toward $1-r$ and checkout-price pass-through becomes weaker. If $r' > 0$, the unreclaimed tax lowers the return rate and shrinks the hit to the effective margin below $1-r$. More complete reclaim shrinks that channel, so the factor rises toward $1-r$.

The companion note also signs the classical bias for ρ_c: the no-return formula overestimates pass-through if and only if $\mu_p > 1$. That comparison does not involve α. At $\tau = 0$ the list-price slope is $\mu_p = 1 - r(p) - r'(p)(p+h)$ under either remittance regime, because reclaim enters only through terms in $\alpha\tau$ that drop out of $\partial\mu/\partial p$. Production-cost pass-through is $\rho_c = S'(\mu)/(S'(\mu)\mu_p - D'(p))$, again with no α. The classical formula sets $\mu_p = 1$ in that expression, so whether it over- or underestimates ρ_c is pinned entirely by whether $\mu_p \gtrless 1$—the same $\varepsilon_r(p) < -p/(p+h)$ cut as in the companion note, for every $\alpha \in [0,1]$. Full reclaim is no exception: equalizing the tax rates leaves ρ_c and its bias untouched.

Result 3 (Pass-through bias under partial reclaim)**.** *The companion bias cut for ρ_c relative to the classical no-return formula is unchanged by α, including at full reclaim $\alpha = 1$. Tax pass-through inherits that bias only through the reclaim factors in* (8)*.*

5 Revenue and first-order surplus

On each gross sale the treasury collects τ. On each return it refunds $\alpha\tau$ to the remitter. Net revenue is therefore

$$G = \tau\, Q\big(1 - \alpha\, r^*\big), \tag{9}$$

where $r^* = r(p)$ under seller remittance and $r^* = r\big(p-(1-\alpha)\tau\big)$ under buyer remittance. At $\tau = 0$ both give $dG/d\tau = Q\big(1-\alpha r(p)\big)$.

Private surplus $W = CS + PS$ is differentiated as in the companion note, using $dCS/d\tau = -Q(1-r)\rho_\tau$ and $dPS/d\tau = Q(\mu_p\rho_\tau + \mu_\tau)$ at $\tau = 0$. Substituting (7)–(8) and writing $\Lambda \equiv 1 + r'(p)(p+h)\rho_c$ yields

$$\frac{dW}{d\tau^s} = -Q\big(1-\alpha r(p)\big)\,\Lambda, \qquad \frac{dW}{d\tau^b} = -Q\big(1 - r(p) - (1-\alpha)\,r'(p)(p+h)\big)\,\Lambda. \tag{10}$$

Social surplus $V = W + G$ then satisfies, at $\tau = 0$,

$$\frac{dV}{d\tau^s} = -Q\big(1-\alpha r(p)\big)\,(\Lambda - 1), \quad \frac{dV}{d\tau^b} = Q\Big[\big(1-\alpha r(p)\big) - \big(1 - r(p) - (1-\alpha)\,r'(p)(p+h)\big)\,\Lambda\Big]. \tag{11}$$

When $\alpha = 0$, (10)–(11) reproduce the companion note. When $\alpha \to 1$, seller remittance has $dW/d\tau^s \to -Q(1-r)\Lambda$ and $dV/d\tau^s \to -Q(1-r)(\Lambda-1)$, so the unclaimed tax on returned units vanishes from the first-order social wedge and only the kept-unit share $1-r$ remains in the private loss. Incomplete reclaim $\alpha < 1$ leaves a strictly larger seller-side resource cost $1 - \alpha r > 1 - r$ whenever $r \in (0,1)$.

Result 4 (Partial recovery nests zero and full reclaim)**.** *For $\alpha \in [0, 1]$, equations* (1)–(11) *nest the zero-reclaim companion note at $\alpha = 0$. At $\alpha = 1$, tax neutrality is restored and the seller-tax pass-through factor collapses to the keep rate $1 - r(p)$.*